Feature 1 — AI Competency & Personalized Learning

Short implementation and working architecture — integrated with Feature 2

1. Purpose

Feature 1 assesses an official's current competencies, compares them with the competencies required for the official's role, identifies and prioritizes skill gaps, recommends relevant iGOT and NSSTA/TPAC training, creates a personalized learning path, and measures improvement through re-assessment.

2. How Feature 1 Works

1	**Official Profile**	Role, designation, department, experience and job activities.
2	**Role & Competency Engine**	Maps the role and job activities to required competencies and levels.
3	**AI Competency Assessment**	Uses Feature 2's competency- and difficulty-tagged MCQs/scenarios.
4	**Competency Scoring**	Calculates the official's current competency from assessment responses.
5	**AI Skill-Gap Analysis**	Compares required vs current competency and identifies gap severity/priority.
6	**Recommendation Engine**	Finds learning resources that can close prioritized gaps.
7	**Personalized Learning Path**	Arranges iGOT and NSSTA/TPAC resources into a learning sequence.
8	**Training Progress**	Tracks course/module progress, completion and assessment activity.
9	**Re-assessment**	Assesses the official again after learning.
10	**Competency Improvement**	Compares before vs after and updates the competency profile.

3. System Architecture

OFFICIAL Profile + Job Activities	**ROLE & COMPETENCY ENGINE** Required Skills + Levels	**FEATURE 2** AI Assessment MCQs / Scenarios
→	→	
COMPETENCY SCORING Current Level	**AI SKILL-GAP ANALYSIS** Required vs Current	**RECOMMENDATION ENGINE** Prioritized Gaps
→	→	
iGOT Courses / Modules	**NSSTA / TPAC** Training Programmes	**PERSONALIZED LEARNING PATH** Ordered Learning
	↓	
TRAINING PROGRESS Completion + Assessment	**RE-ASSESSMENT** Before vs After	**COMPETENCY IMPROVEMENT** Updated Profile

The central loop is: **Official Profile** → **Required Competencies** → **Feature 2 Assessment** → **Current Competency** → **Skill Gap** → **Learning** → **Re-assessment** → **Updated Profile.**

4. Feature 2 Integration

Feature 2 creates the assessment material: learning material is processed, AI generates structured MCQs, questions are validated and tagged with competency and difficulty. Feature 1 consumes those questions, records answers, scores competencies and uses the results for skill-gap analysis. Feature 2 can send Feature 1 a structured result such as **user_id + competency + score + assessment_type**.

5. Step-by-Step Working of Feature 1

What happens at each stage, why it is needed, and the expected output.

Step 1 — Official Profile

The official enters role information such as designation, department, experience and job activities. This gives the system the context needed to determine which competencies are relevant.

Output: Official profile + job activities.

Step 2 — Role & Competency Engine

The system maps the official's role and activities to required competencies. For each competency it stores a target proficiency level, for example Survey Methodology = Level 4 and Data Visualization = Level 3.

Output: Required competency list with target levels.

Step 3 — AI Competency Assessment

Feature 1 requests suitable questions from Feature 2. The questions already contain competency and difficulty tags. The official answers MCQs/scenarios, and the assessment can move from Easy → Medium → Hard or the reverse based on performance.

Output: Assessment responses linked to competencies.

Step 4 — Competency Scoring

The system groups assessment responses by competency and calculates performance. A simple MVP can convert the percentage into levels, for example 0–40% = Level 1, 41–60% = Level 2, 61–80% = Level 3, and 81–100% = Level 4.

Output: Current competency score/level.

Step 5 — AI Skill-Gap Analysis

The system compares required competency with current competency. The basic calculation is **Gap = Required Level** − **Current Level**. It then identifies which competencies have the largest or most important gaps.

Output: Gap size, severity and priority.

Step 6 — Recommendation Engine

For each prioritized gap, the system searches a verified learning catalogue. It should match resources to the competency, current level and required level rather than simply showing a generic course list.

Output: Ranked learning recommendations.

Step 7 — Personalized Learning Path

The recommendations are arranged into a sequence. For example: Survey Methodology Basics → Sampling Techniques → Advanced Survey Design → NSSTA/TPAC Programme → Assessment.

Output: Ordered learning path.

Step 8 — Training Progress

The system tracks whether a learning resource has been started, how much has been completed, whether it is finished, and any related assessment activity.

Output: Training progress and completion status.

Step 9 — Re-assessment

After training, the official takes another competency assessment using relevant questions. The new result is compared with the earlier assessment.

Output: Post-training competency result.

Step 10 — Competency Improvement

The system calculates the improvement, updates the official's competency profile and checks for any remaining gap. If a gap remains, another learning resource can be recommended. This makes Feature 1 a continuous learning loop.

Output: Updated competency profile + remaining gaps.

6. Main Components

Component	Role
React Frontend	Profile form, assessment UI, dashboard, learning path and progress.
Node.js + Express	APIs, scoring, gap analysis, recommendations and Feature 2 communication.
Feature 2	Generates/validates assessment questions and provides adaptive assessment.

Supabase	Stores profiles, roles, competencies, responses, results, resources and progress.
Recommendation Engine	Matches prioritized competency gaps with verified iGOT and NSSTA/TPAC resources.

7. Recommended MVP Build Order

1. Competency database → **2.** Official profile → **3.** Feature 2 integration → **4.** Answer storage → **5.** Competency scoring → **6.** Gap analysis → **7.** iGOT/NSSTA-TPAC catalogue → **8.** Recommendation engine → **9.** Dashboard → **10.** Re-assessment.

Final result: Feature 1 is not only an assessment screen. It creates a continuous cycle of assessment → competency profile → skill-gap analysis → targeted training → re-assessment → measurable improvement.